Software Requirements Specification

Hostel Management System (HMS)

Team 1

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Revision history

Version	Date	Author	Change summary	Approval
1.0	09-09-2026	Team 1	Initial SRS for the Hostel Management System, with use-case diagrams and RTM	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification for a Hostel Management System (HMS) intended for use by an educational institution. It defines the functional and non-functional requirements, the external interfaces and the verification criteria for software that manages hostel admission, room allocation, fee collection, gate and outpass control, mess operations, maintenance complaints and administrative reporting.

1.2 Scope

The system covers the full residency life cycle of a student: application, room allocation, fee payment, day-to-day residency (attendance, outpass, mess, complaints, visitors) and vacating with no-dues clearance. It integrates with the institutional student-records system for identity verification, with a payment gateway for online collections, and with an SMS or email service for notifications.

Excluded from scope: the internal accounting ledger of the institution beyond the invoice and receipt interface, academic examination or attendance records, payroll for hostel staff, and physical security hardware other than the interface to the existing RFID or biometric gate reader.

1.3 Audience

Developers, QA engineers, hostel wardens and administrative staff, the mess contractor, and assessment evaluators.

1.4 Definitions and acronyms

Term	Meaning
HMS	Hostel Management System
Resident	A student to whom a bed has been allocated
Warden	The staff member responsible for a hostel block
Outpass	An approved permission for a resident to leave the hostel for a stated period
Curfew	The configured time by which residents must return
No-dues	Confirmation that no fee, mess or damage amount is outstanding
Bed	The smallest allocatable unit; a room contains one or more beds
RFID	Radio-Frequency Identification, used for gate entry cards
SLA	Service Level Agreement, the time limit for resolving a complaint
RTM	Requirements Traceability Matrix
RTO / RPO	Recovery Time Objective / Recovery Point Objective
MFA	Multi-Factor Authentication

2. Overall description

2.1 Product perspective

The HMS is a new, self-contained web application with a responsive resident portal and a staff back-office. It replaces the spreadsheets and paper registers currently used by the hostel office. It reads student identity data from the institutional ERP, delegates payment capture to an external gateway, sends messages through an SMS or email provider, and reads entry events from the existing RFID or biometric gate reader. The application follows a three-tier architecture: browser client, application server exposing REST services, and a relational database.

2.2 Major product functions

- Hostel admission application and approval
- Automatic and manual room and bed allocation, transfer and vacating
- Fee invoicing, online payment, receipting and overdue tracking
- Gate entry-exit capture, outpass workflow, curfew alerts and night attendance
- Mess menu publication, opt-out handling and consumption-based billing
- Maintenance complaint lodging, assignment, SLA tracking and escalation
- Visitor registration and visiting-hours enforcement
- Occupancy, dues, complaint-ageing and mess reports with export
- User, role and master-data administration with audit logging

2.3 User roles and characteristics

Role	Characteristics
Resident (student)	Everyday smartphone user; uses the portal a few times a month; expects a simple mobile-friendly interface.
Warden	Moderate computer skill; uses the system daily for approvals, allocation and attendance; needs dashboards rather than deep menus.
Security guard	Low computer skill; operates only the gate and visitor screens, often in a hurry; needs large controls and minimal typing.
Maintenance staff	Low to moderate skill; receives and closes assigned complaints, usually on a mobile device.
Mess manager	Moderate skill; maintains menus and reviews consumption.
Hostel administrator	Advanced user; manages masters, users, roles, fee structure and reports.
Guardian	Not a system user; receives notifications only.

2.4 Operating environment

Server side: Linux application server with a relational database (PostgreSQL or MySQL), hosted in the institutional data centre or on a cloud VM. Client side: current versions of Chrome, Edge, Firefox and Safari on desktop, and Android and iOS mobile browsers. The gate terminal runs on a networked workstation attached to the RFID or biometric reader. The system operates over the campus LAN with internet access for the payment gateway and notification service.

2.5 Constraints and assumptions

- Card payments must be handled by a PCI-DSS compliant gateway; card data is never stored by the HMS.
- Personal data of residents, several of whom may be minors, must be handled in line with applicable data-protection rules and institutional policy.
- Hostels are gender-segregated; allocation logic must never place a resident in a block of the opposite gender.
- Student identity is authoritative in the institutional ERP; the HMS does not create student master records.

- The gate reader vendor exposes a documented event API; changing the vendor would require an adapter change.
- Network connectivity at the gate terminal may be intermittent; entry events must be buffered locally and synchronised when the link returns.

3. External interface requirements

3.1 User interfaces

Resident portal: responsive web pages for application, allocation status, fees and receipts, outpass requests, mess opt-out and complaints. Staff back-office: dashboard-led screens for allocation, approvals, attendance, complaints and reports. Gate terminal: a single high-contrast screen showing the scanned resident's photo, name, room and current IN or OUT state, operable with minimal typing. All screens conform to WCAG 2.1 Level AA and support keyboard navigation.

3.2 Hardware interfaces

- RFID card reader or fingerprint scanner at the hostel gate (vendor event API over the local network)
- Gate terminal workstation with display and optional buzzer for violation alerts
- Network printer for receipts, allocation letters and attendance registers
- Standard desktop and mobile devices for staff and residents

3.3 Software interfaces

Interface	Purpose
Institutional ERP / student records API	Read-only lookup of enrolment number, name, programme and photograph, over HTTPS with JSON.
Payment gateway API	Payment initiation, callback and refund, over HTTPS with signed requests.
SMS and email notification service	Templated messages to residents, guardians and staff.
Relational database	PostgreSQL or MySQL, accessed only through the application's data layer.
Reporting and export library	PDF and XLSX generation for receipts and reports.

3.4 Communications interfaces

- HTTPS with TLS 1.2 or higher for all browser and third-party traffic.
- REST over JSON for internal service calls, with token-based authentication.
- Retry with exponential backoff for gateway and notification failures, and a dead-letter queue for messages that still fail after three attempts.
- Local buffering of gate events during network loss, with ordered replay on reconnection.

4. System features (detailed)

Each requirement below carries an acceptance criterion and a reference test case. Functional IDs follow the pattern HMS-F-###. Twenty functional requirements are specified across seven feature areas.

4.1 Admission and Room Allocation

Description: *Capture hostel applications, verify eligibility against student records, and allocate beds subject to gender, block and preference rules.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-001	The system shall allow an applicant to submit a hostel admission application capturing personal, academic, guardian and medical details, and shall reject a duplicate application for the same enrolment number in the same academic session.	Functional	High	Hostel Office	AC-HMS-F-001: An application number is issued only when all mandatory fields pass validation; a second application for the same enrolment number is rejected with an explanatory message. Test: TC-ADM-01	Requires student master data from the college ERP; enrolment number is the natural key
HMS-F-002	The system shall allocate a bed automatically from vacant beds matching the applicant's gender, requested block and room type, following the merit or seniority order configured by the administrator.	Functional	High	Warden	AC-HMS-F-002: Allocation is offered only from vacant, gender-matching beds and follows the configured ranking. Test: TC-ALL-01	Depends on room inventory master (HMS-F-061)
HMS-F-003	The system shall let the warden reallocate or transfer a resident to another bed, recording reason, actor and timestamp.	Functional	Medium	Warden	AC-HMS-F-003: Source bed becomes vacant and target bed occupied in a single transaction; an audit entry is written. Test: TC-ALL-02	Audit log (HMS-SR-004)

4.2 Fee and Payment Management

Description: *Raise invoices on allocation, collect payment online, issue receipts and track overdue amounts.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-010	The system shall generate an itemised fee invoice (room rent, mess advance, security deposit, maintenance) when an allocation is confirmed.	Functional	High	Accounts	AC-HMS-F-010: Invoice total equals the sum of configured fee heads for the resident's room type. Test: TC-FEE-01	Fee structure master must be configured
HMS-F-011	The system shall accept online fee payment through the integrated payment gateway and record the transaction reference and status against the invoice.	Functional	High	Accounts / Resident	AC-HMS-F-011: A successful gateway callback marks the invoice Paid; a failed or abandoned payment leaves it Unpaid. Test: TC-FEE-02	Payment gateway API; see HMS-SR-005
HMS-F-012	The system shall issue a system-generated receipt with a unique receipt number, downloadable as PDF, within 30 seconds of a successful payment.	Functional	High	Accounts	AC-HMS-F-012: Receipt number is unique and the PDF is available in the resident portal. Test: TC-FEE-03	PDF rendering service
HMS-F-013	The system shall mark invoices unpaid after the due date as Overdue and send automated reminders to the resident and the guardian.	Functional	Medium	Accounts	AC-HMS-F-013: Status changes at 00:00 on the day after the due date and a reminder is dispatched. Test: TC-FEE-04	Notification service

4.3 Attendance, Outpass and Gate Management

Description: *Record movement at the gate, manage leave approvals, alert on curfew violations and produce the night-attendance register.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-020	The system shall record every entry and exit at the hostel gate through the RFID or biometric reader, with a manual entry fallback for the security guard.	Functional	High	Security / Warden	AC-HMS-F-020: Each scan creates a timestamped IN or OUT record against the resident. Test: TC-GATE-01	RFID/biometric device SDK
HMS-F-021	The system shall allow a resident to raise an outpass or leave request stating destination, contact number and expected return date-time, routed to the warden for approval, and shall notify the registered guardian when the request is approved and when the resident is recorded as returned.	Functional	High	Resident / Warden / Guardian	AC-HMS-F-021: The request state moves through Pending, Approved or Rejected, and both guardian notifications are dispatched. Test: TC-OUT-01	Workflow engine; notification service; guardian consent
HMS-F-022	The system shall raise a curfew-violation alert to the warden when a resident is not recorded as returned within 30 minutes of the configured curfew time.	Functional	High	Warden	AC-HMS-F-022: Alert appears on the warden dashboard and is pushed within 5 minutes of the threshold. Test: TC-GATE-02	Curfew time configurable per hostel
HMS-F-023	The system shall produce a daily night-attendance register listing residents as Present, On-leave or Absent, room by room.	Functional	Medium	Warden	AC-HMS-F-023: Register is generated after curfew and matches gate and outpass records. Test: TC-ATT-01	Derived from HMS-F-020 and HMS-F-021

4.4 Mess Management

Description: *Publish the menu, handle opt-outs and bill on consumed days.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-030	The system shall publish the weekly mess menu maintained by the mess manager and make it visible to all residents.	Functional	Low	Mess Manager	AC-HMS-F-030: Published menu is visible to residents within 1 minute of saving. Test: TC-MESS-01	Menu is versioned by week
HMS-F-031	The system shall allow a resident to opt out of mess for a date range given at least 24 hours' notice, and shall compute the monthly mess bill on consumed days only.	Functional	High	Mess Manager / Accounts	AC-HMS-F-031: Opt-out within 24 hours is refused; bill equals rate multiplied by consumed days. Test: TC-MESS-02	Feeds fee module (HMS-F-010)

4.5 Complaint and Maintenance Management

Description: *Lodge, assign, track and escalate maintenance complaints against defined service levels.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-040	The system shall allow a resident to lodge a maintenance complaint with a category, description and optional photograph, and shall assign a priority from the category.	Functional	High	Residents	AC-HMS-F-040: A ticket ID is returned and the complaint appears in the maintenance queue. Test: TC-CMP-01	Image upload limited to 5 MB
HMS-F-041	The system shall let the warden assign a complaint to maintenance staff and track it through the states Open, In-progress, Resolved and Closed, with closure confirmed by the complainant.	Functional	High	Warden / Maintenance	AC-HMS-F-041: State transitions follow the defined order and closure requires complainant confirmation. Test: TC-CMP-02	Workflow engine
HMS-F-042	The system shall escalate a complaint to the hostel administrator when it is unresolved beyond its SLA (48 hours for High, 96 hours for Medium).	Functional	Medium	Administration	AC-HMS-F-042: An escalation notice is generated at SLA breach and logged against the ticket. Test: TC-CMP-03	SLA values are configurable

4.6 Visitor Management

Description: *Register visitors at the gate and enforce visiting hours.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-050	The system shall register every visitor with name, relationship to the resident, identity-proof number and entry and exit timestamps, and shall block entry outside configured visiting hours unless the warden authorises it with a recorded reason.	Functional	Medium	Security Guard / Warden	AC-HMS-F-050: No visitor record can be closed without an exit timestamp; out-of-hours entry is refused without a warden override carrying a reason. Test: TC-VIS-01	ID proof number stored encrypted; override is an auditable event

4.7 Administration and Reporting

Description: *Maintain masters, manage the vacating process and generate management reports.*

Req ID	Requirement	Type	Priority	Source	Acceptance criteria / Test	Comments
HMS-F-060	The system shall generate occupancy (total, occupied, vacant and blocked beds per block and floor), outstanding-dues, complaint-ageing and mess-consumption reports for a chosen period, exportable to PDF and XLSX.	Functional	Medium	Administration	AC-HMS-F-060: Each report reconciles with source records and both export formats open correctly. Test: TC-RPT-01	Reporting library
HMS-F-061	The system shall allow the administrator to maintain the room inventory, including block, floor, room number, room type, bed capacity and status.	Functional	High	Administration	AC-HMS-F-061: A room cannot be deleted while any bed in it is occupied. Test: TC-INV-01	Master data for HMS-F-002
HMS-F-062	The system shall process a vacating request only after no-dues clearance is recorded by the fee, mess and maintenance modules, shall then release the bed to the vacant pool, and shall compute the refundable security deposit after deducting outstanding dues and recorded damage charges.	Functional	High	Accounts / Warden	AC-HMS-F-062: The bed is released only when all three clearances are present; refund equals deposit minus dues and damages, a negative result being shown as payable. Test: TC-VAC-01	Depends on HMS-F-010 and HMS-F-031

5. Non-functional requirements (detailed)

The requirements below are measurable and tied to the test plan. IDs follow the pattern HMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria
HMS-NF-001	The system shall return any resident-portal page or search result within 2 seconds for 95% of requests, and shall support at least 2,000 resident records and 200 concurrent users without exceeding that target.	Performance / Scalability	High	95th percentile $\leq$ 2 s in a production-like load test at 200 concurrent users with 2,000 seeded records. Test: TC-PERF-01
HMS-NF-002	The system shall be available 99.5% of the time during the academic session, excluding announced maintenance windows.	Reliability / Availability	High	Monthly uptime reports show $\geq$ 99.5%. Test: operations monitoring reports
HMS-NF-003	The system shall take a full database backup nightly, with a recovery point objective of 24 hours and a recovery time objective of 4 hours.	Recoverability / Backup	High	A quarterly restore drill completes within 4 hours with no data loss beyond 24 hours. Test: TC-OPS-01
HMS-NF-004	A warden with no prior training shall be able to complete a room allocation in under 5 minutes using on-screen guidance, and the resident portal shall conform to WCAG 2.1 Level AA and render correctly on mobile screens 360 px wide and above.	Usability / Accessibility	Medium	80% of a 10-user trial group complete the allocation unaided within 5 minutes; accessibility audit passes with no Level AA violations. Test: TC-UX-01
HMS-NF-005	The system shall retain resident, fee and gate-entry records for 5 years, after which personal data shall be archived or anonymised.	Audit / Data retention	Medium	Archival and retrieval of a 5-year-old record verified; anonymisation job runs on schedule. Test: TC-OPS-02

5.1 Security

5.1.1 Security objectives

The objectives below state what the system must protect and why. They are the basis for the security requirements in 5.1.2.

ID	Objective	Rationale
SO-1	Confidentiality of resident data	Personal, guardian, medical and identity-proof details — including those of residents who are minors — must be visible only to the roles that need them, and never to other residents.
SO-2	Integrity and non-repudiation of financial and allocation records	Every fee transaction, allocation, transfer and approval must be attributable to an identified actor and must not be alterable without an audit trail. No single account may both approve an allocation or waiver and alter the audit record of that approval.
SO-3	Controlled physical access	Gate, outpass and visitor records must be accurate and timely, so that the hostel always knows who is inside the premises.

5.1.2 Security requirements

Req ID	Requirement	Type	Priority	Acceptance criteria / Test
HMS-SR-001	The system shall enforce TLS 1.2 or higher on all network connections, including browser, payment-gateway and notification traffic.	Security	High	TLS scan reports no connection below TLS 1.2 and no weak ciphers. Test: TC-SEC-01
HMS-SR-002	The system shall enforce role-based access control on every function, granting each role only the least privilege required, and shall terminate an authenticated session after 15 minutes of inactivity and invalidate the session token on logout.	Security	High	Each role is tested against every restricted endpoint and unauthorised access returns HTTP 403; a session idle for 15 minutes and a token reused after logout are both rejected. Test: TC-SEC-02
HMS-SR-003	The system shall store passwords only as salted hashes using bcrypt or Argon2, and shall require multi-factor authentication for Warden and Administrator accounts.	Security	High	Database inspection shows no reversible password; MFA challenge is mandatory for privileged logins. Test: TC-SEC-03
HMS-SR-004	The system shall write a tamper-evident audit log entry for every allocation, fee, outpass approval, override and account-management action, recording actor, action, timestamp and outcome.	Security / Audit	High	A sample of each action type appears in the log; entries are append-only and cannot be edited from the application. Test: TC-SEC-04
HMS-SR-005	The system shall process payments only through a PCI-DSS compliant gateway and shall never store card numbers, CVV or bank credentials in its own database.	Security / Compliance	High	Database and log review confirms no cardholder data is retained. Test: TC-SEC-05
HMS-SR-006	The system shall validate and sanitise all user input and use parameterised queries to prevent SQL injection and cross-site scripting.	Security	High	OWASP ZAP scan reports no High or Medium injection or XSS findings. Test: TC-SEC-06

6. Quality attributes & Acceptance tests

Exit criteria for acceptance: every High-priority functional requirement is implemented and verified, no non-functional or security requirement fails, all High-severity defects are closed, and the RTM shows a Pass status for every listed test case.

Acceptance test suites:

Suite	Covers	Test cases
TS-1	Admission and Allocation	TC-ADM-01, TC-ALL-01, TC-ALL-02, TC-INV-01
TS-2	Fees and Payments	TC-FEE-01 to TC-FEE-04
TS-3	Gate, Outpass and Attendance	TC-GATE-01, TC-GATE-02, TC-OUT-01, TC-ATT-01
TS-4	Mess	TC-MESS-01, TC-MESS-02
TS-5	Complaints	TC-CMP-01 to TC-CMP-03
TS-6	Visitors and Vacating	TC-VIS-01, TC-VAC-01
TS-7	Reporting and Operations	TC-RPT-01, TC-PERF-01, TC-OPS-01, TC-OPS-02
TS-8	Security	TC-SEC-01 to TC-SEC-06
TS-9	Usability and Accessibility	TC-UX-01

Quality attributes summary: performance and scalability are verified by load testing (HMS-NF-001); reliability and recoverability by uptime reporting and a quarterly restore drill (HMS-NF-002, HMS-NF-003); usability and accessibility by a supervised user trial and an accessibility audit (HMS-NF-004); and security by the test cases attached to HMS-SR-001 to HMS-SR-006 together with an OWASP-based vulnerability scan.

7. System models and diagrams

7.1 UML use-case diagrams

Two use-case diagrams are provided. Figure 1 shows the services available to the resident and the external actors they involve. Figure 2 shows the warden, security, maintenance and administration side of the same system. The «include» relationships mark behaviour that is always invoked as part of the base use case.

Primary actors. Resident, Warden, Security Guard, Maintenance Staff, Mess Manager, Hostel Administrator.

Secondary and external actors. Payment Gateway, Notification Service, Guardian.

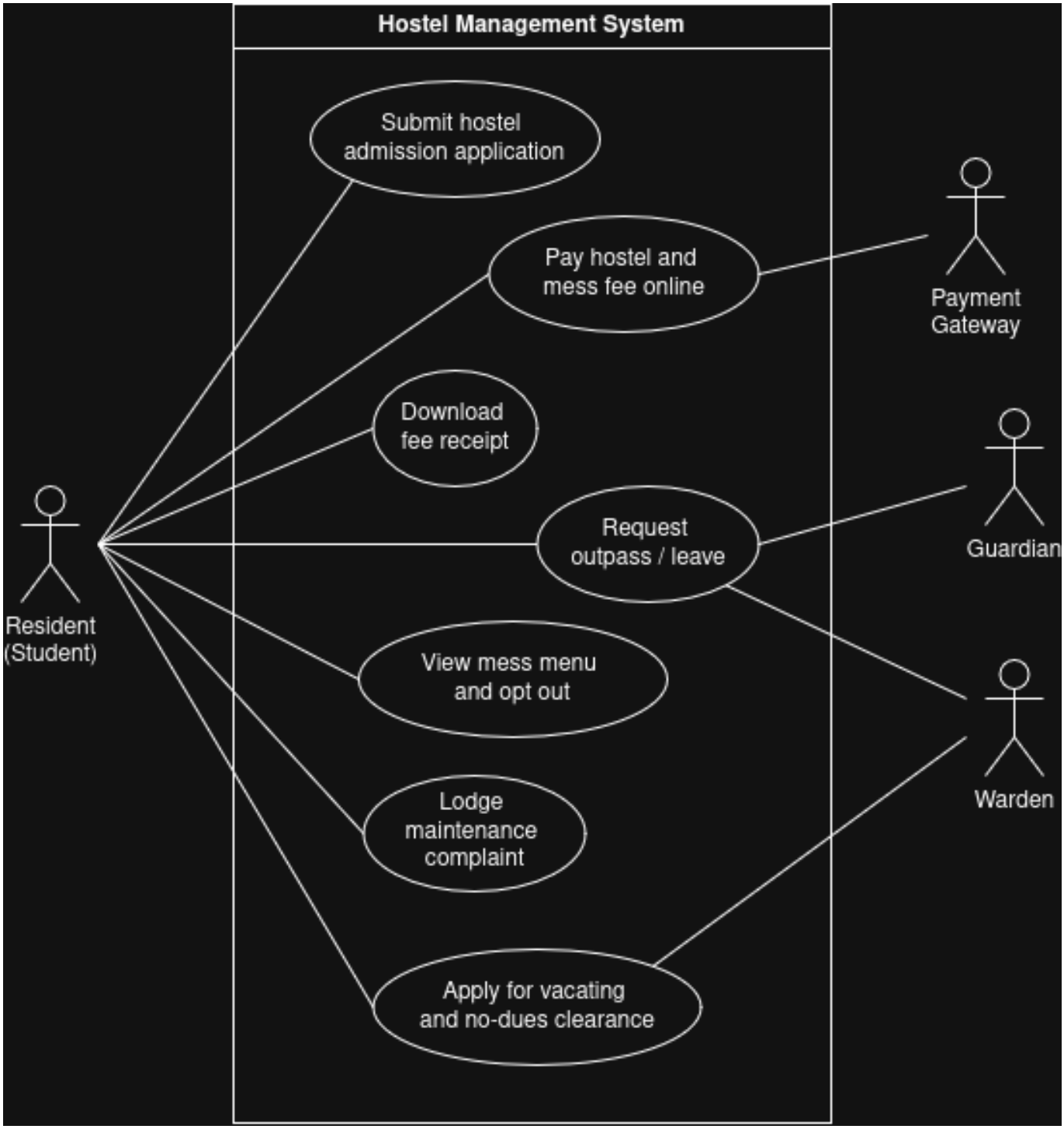


Figure 1 — Use-case diagram: resident-facing services

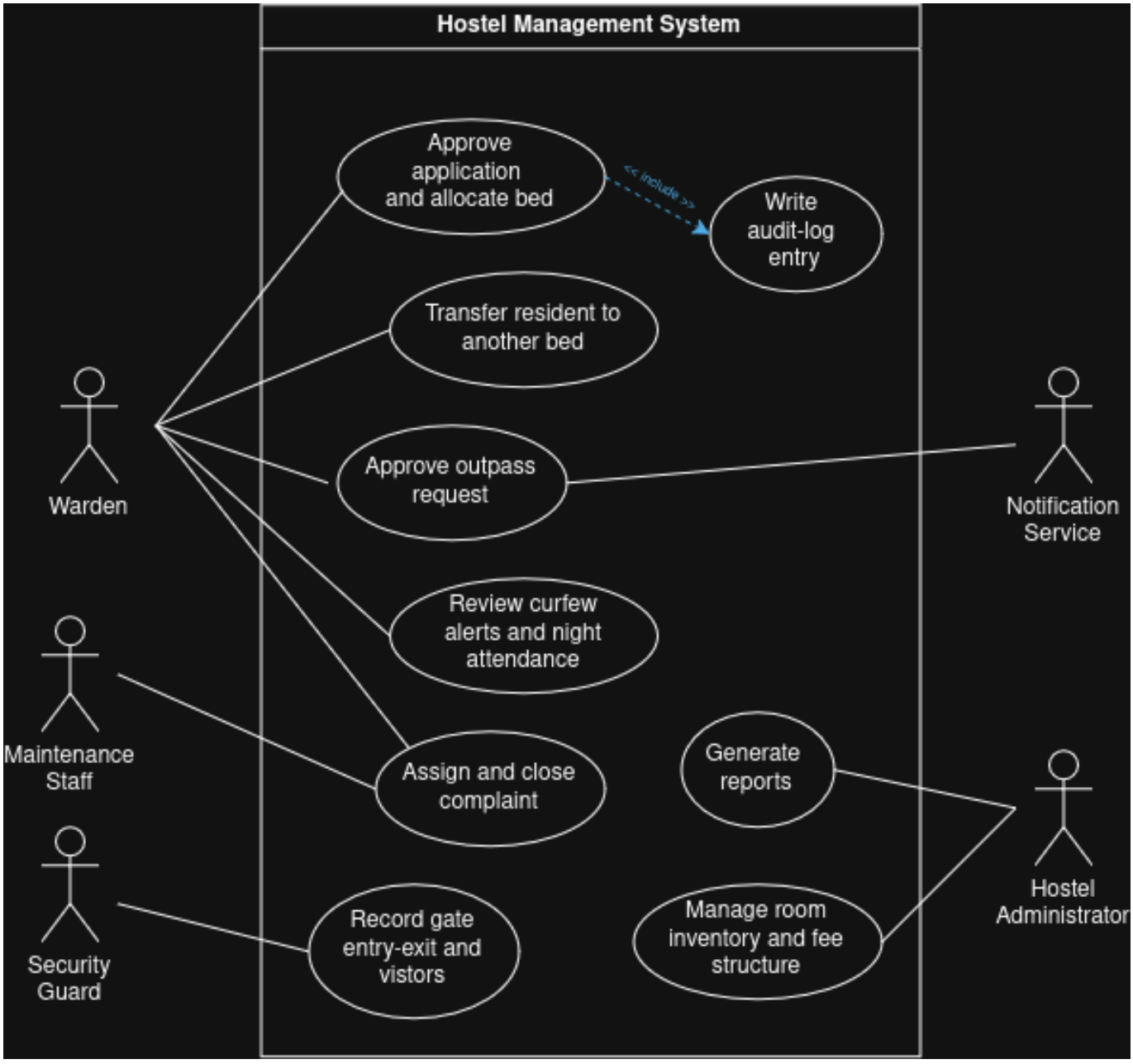


Figure 2 — Use-case diagram: warden, staff and administration

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement	Section / Design spec	Module	Test cases	Status	Comments
HMS-F-001	Submit admission application	4.1 / DS-ADM-01	AdmissionModule	TC-ADM-01	A	
HMS-F-002	Automatic bed allocation	4.1 / DS-ALL-01	AllocationModule	TC-ALL-01	A	Core feature
HMS-F-003	Room transfer by warden	4.1 / DS-ALL-02	AllocationModule	TC-ALL-02	A	Dev. 5
HMS-F-010	Generate fee invoice	4.2 / DS-FEE-01	FeeModule	TC-FEE-01	A	
HMS-F-011	Online fee payment	4.2 / DS-FEE-02	FeeModule / Gateway	TC-FEE-02	A	External dependency; Dev. 3
HMS-F-012	Receipt generation	4.2 / DS-FEE-03	FeeModule	TC-FEE-03	A	
HMS-F-013	Overdue flag and reminders	4.2 / DS-FEE-04	FeeModule / Notify	TC-FEE-04	A	Dev. 4, 7
HMS-F-020	Gate entry-exit capture	4.3 / DS-GATE-01	GateModule	TC-GATE-01	A	Needs device SDK; Dev. 8, 15
HMS-F-021	Outpass workflow and guardian notice	4.3 / DS-OUT-01	OutpassModule	TC-OUT-01	A	Dev. 7
HMS-F-022	Curfew violation alert	4.3 / DS-GATE-02	GateModule	TC-GATE-02	A	Dev. 9
HMS-F-023	Night-attendance register	4.3 / DS-ATT-01	AttendanceModule	TC-ATT-01	A	
HMS-F-030	Publish mess menu	4.4 / DS-MESS-01	MessModule	TC-MESS-01	A	Dev. 10
HMS-F-031	Mess opt-out and billing	4.4 / DS-MESS-02	MessModule	TC-MESS-02	A	Dev. 10
HMS-F-040	Lodge complaint	4.5 / DS-CMP-01	ComplaintModule	TC-CMP-01	A	
HMS-F-041	Complaint lifecycle	4.5 / DS-CMP-02	ComplaintModule	TC-CMP-02	A	
HMS-F-042	SLA escalation	4.5 / DS-CMP-03	ComplaintModule	TC-CMP-03	A	Dev. 11
HMS-F-050	Visitor registration and hours	4.6 / DS-VIS-01	VisitorModule	TC-VIS-01	A	Dev. 12
HMS-F-060	Management reports	4.7 / DS-RPT-01	ReportModule	TC-RPT-01	A	Dev. 13
HMS-F-061	Room inventory master	4.7 / DS-INV-01	AdminModule	TC-INV-01	A	
HMS-F-062	No-dues clearance and refund	4.7 / DS-VAC-01	VacateModule	TC-VAC-01	A	
HMS-NF-001	Response time and capacity	5 / DS-PERF-01	Portal / API	TC-PERF-01	P	Partial: p95 1.6 s on a laptop, not a production-like server (Dev. 14)
HMS-NF-002	Availability 99.5%	5 / DS-OPS-01	Infrastructure	Ops reports	P	Partial: uptime script built; a month of data not collected (Dev. 14)
HMS-NF-003	Backup and recovery	5 / DS-OPS-02	Infrastructure	TC-OPS-01	A	Dev. 14
HMS-NF-004	Usability and accessibility	5 / DS-UX-01	Portal	TC-UX-01	P	Partial: 0 axe violations on 33 pages; no 10-user trial (Dev. 17)
HMS-NF-005	Data retention	5 / DS-OPS-03	Infrastructure / DB	TC-OPS-02	A	
HMS-SR-001	TLS 1.2 or higher	5.1.2 / DS-SEC-01	Infrastructure	TC-SEC-01	P	SO-1; Partial: enforced and tested in the app; no live TLS scan (Dev. 6, 17)
HMS-SR-002	RBAC and session expiry	5.1.2 / DS-SEC-02	AuthModule	TC-SEC-02	A	SO-1, SO-2
HMS-SR-003	Password hashing and MFA	5.1.2 / DS-SEC-03	AuthModule	TC-SEC-03	A	SO-1
HMS-SR-004	Audit logging	5.1.2 / DS-SEC-04	AuditModule	TC-SEC-04	A	SO-2

Req ID	Requirement	Section / Design spec	Module	Test cases	Status	Comments
HMS-SR-005	No cardholder data stored	5.1.2 / DS-SEC-05	FeeModule / Gateway	TC-SEC-05	A	SO-1
HMS-SR-006	Input validation and injection defence	5.1.2 / DS-SEC-06	All modules	TC-SEC-06	P	SO-2; Partial: injection and markup tests pass; no ZAP scan (Dev. 17)

Status codes: N = Not started, P = Partial (works; some evidence not available), A = Accepted. Status as of the System Validation Report, 8 November 2026. "Dev. n" is row n of DEVIATIONS.md in the project repository.